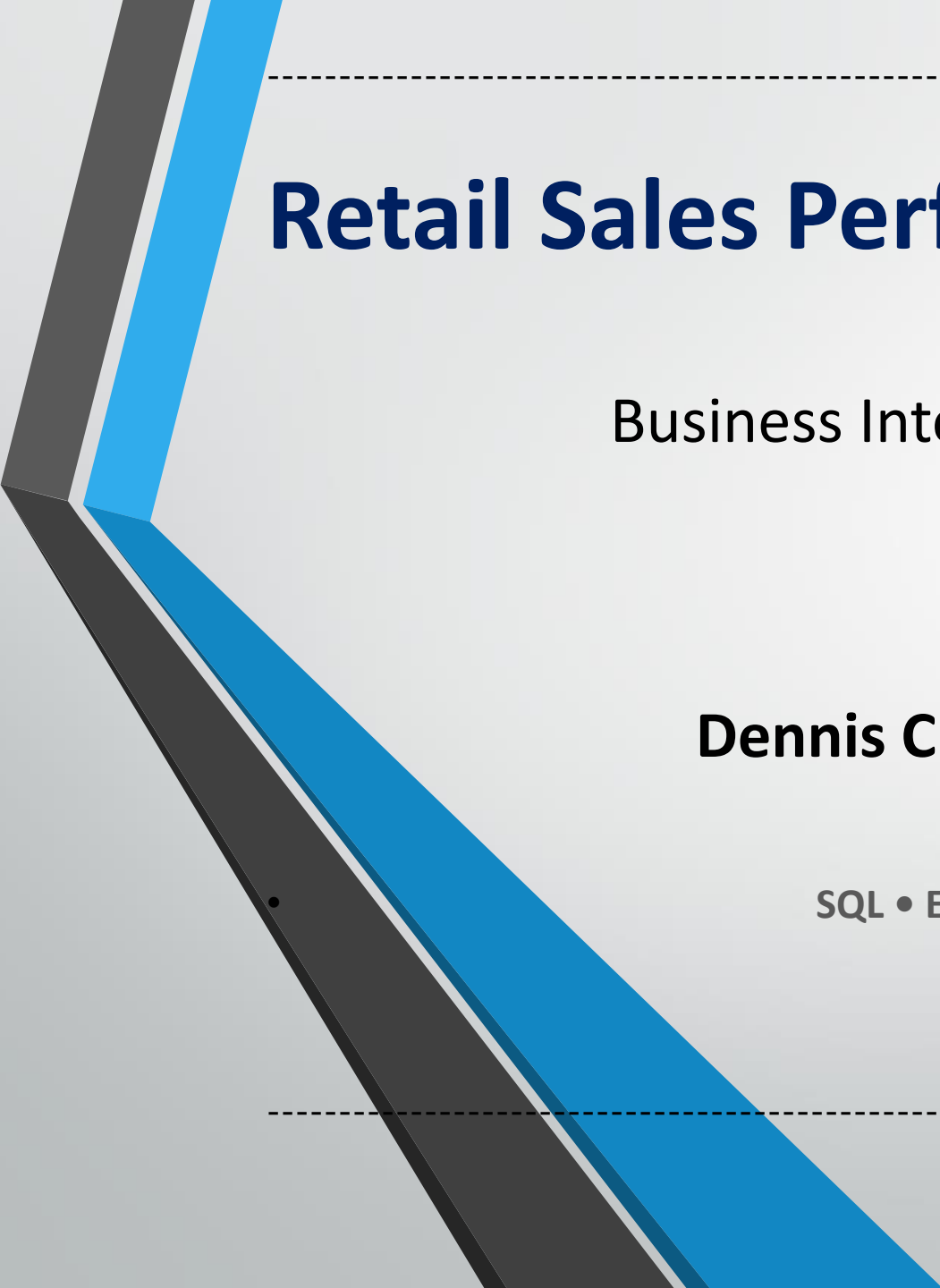




Retail Sales Performance Dashboard

Business Intelligence Portfolio Project

Dennis Chinonso Ugwuonu

SQL • Excel • Power BI • GitHub

July 2026

Business Problem

Business Challenge

- Sales data is collected across products, customers, states, and payment methods.
- The data exists in raw form and is difficult to analyze.
- Management lacks a centralized view of business performance.
- This limits timely, data-driven decision-making.

Project Objective

- Build an interactive Business Intelligence dashboard.
- Transform raw sales data into meaningful insights.
- Monitor KPIs and sales performance.
- Support faster, evidence-based business decisions.

Key Business Questions

- Which **products** and **categories** generate the highest **sales** and **profit**?
- Which **states** contribute the most to **total revenue**?
- How do **monthly sales** and **profits** change over time?
- Who are the company's **top-performing customers**?
- Which **product categories** require improvement in **profitability**?

Dataset & Tools Used

Project Resources and Tech Stack

Dataset

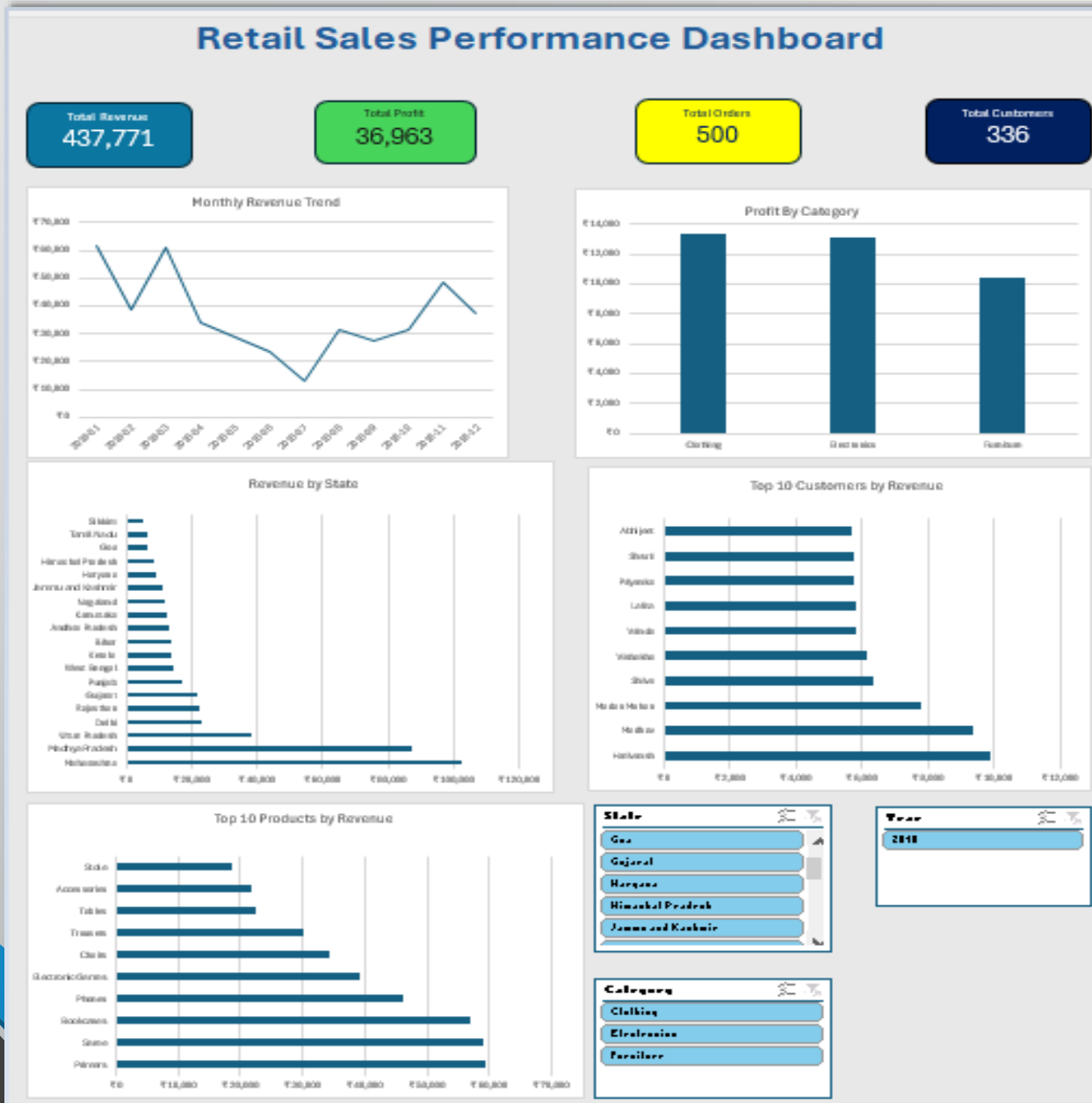
- **Madhav E-Commerce Sales Dataset**
- Contains **orders, customers, products, payment methods, profit, sales, and geographic data.**
- Data was **cleaned and transformed** before analysis.
- Used for **dashboard** development and **KPI** reporting.

Tools Used

- **SQL (DuckDB)** – Data querying and analysis
- **Microsoft Excel** – Data cleaning and KPI analysis
- **Power BI** – Dashboard development and visualization
- **GitHub** – Documentation and portfolio management

Dashboard Overview

Interactive Excel dashboard developed to monitor retail sales performance through key business metrics and visual analytics.



Dashboard Features

- ✓ KPI Summary Cards
- ✓ Monthly Revenue Trend
- ✓ Profit by Category
- ✓ Sales by State
- ✓ Top Customers
- ✓ Top Products
- ✓ Interactive Slicers (State, Category, Year)

Key Insights

Sales Performance

- Tracks KPIs and monthly sales trends.
- Highlights regional performance.
- Supports growth planning.

Customer Insights

- Identifies top customers.
- Reveals purchasing patterns.
- Improves retention strategies.

Product & Category Insights

- Highlights top-selling products.
- Compares profitability.
- Supports pricing decisions.

Regional Performance

- Compares state performance.
- Identifies growth opportunities.
- Supports resource allocation.

These insights support data-driven business decisions and strategic planning.

Business Recommendations

Increase Investment

- Focus on top-selling products.
- Optimize inventory.
- Increase marketing.

Customer Retention

- Reward loyal customers.
- Personalize promotions.
- Encourage repeat purchases.

Regional Performance

- Improve low-performing regions.
- Launch targeted campaigns.
- Optimize distribution.

Data-Driven Decisions

- Monitor KPIs.
- Review dashboard regularly.
- Support strategic planning.

These recommendations help maximize profitability, improve customer loyalty, and support sustainable business growth.

Lessons Learned

Technical Skills Gained

- Improved **SQL** skills for business analysis.
- Strengthened **Microsoft Excel** skills for data cleaning and KPI reporting.
- Built **Pivot Tables** and **Pivot Charts**.
- Developed dashboard design and data visualization skills.
- Improved **GitHub** documentation and presentation skills.

Challenges Overcome

- Combined multiple tables into one analysis dataset.
- Resolved data formatting and cleaning issues.
- Completed the BI dashboard despite Power BI installation challenges.
- Strengthened problem-solving through practical project work.

Future Improvements

- Develop an interactive **Power BI** dashboard.
- Analyze larger and more diverse datasets.
- Add forecasting and advanced KPIs.
- Automate dashboard refresh and reporting.



Thank You!

Questions & Discussions

Presented by

Dennis Chinonso Ugwuonu

Business Intelligence Portfolio Project

[GitHub Portfolio](#)

Thank you for your time and consideration.

Thank You!

Questions & Discussion

Dennis Chinonso Ugwuonu

Retail Sales Performance Dashboard
Business Intelligence Portfolio Project

GitHub: [GitHub Portfolio](#)

Email: dennis.ugwuonu@gmail.com

Thank you for your time and attention.